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1) Configuring a Basic Spring Application

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="
    http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.student</groupId>
  <artifactId>StudentManagement</artifactId>
  <version>1.0</version>

  <dependencies>

    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-context</artifactId>
      <version>6.1.8</version>
    </dependency>

  </dependencies>

</project>
```

StudentRepository.java

```
package com.student.repository;

public class StudentRepository {

  public void displayStudent() {

    System.out.println("Student information fetched successfully.");

  }
}
```

```
}
```

StudentService.java

```
package com.student.service;

import com.student.repository.StudentRepository;

public class StudentService {

    private StudentRepository studentRepository;

    public void setStudentRepository(StudentRepository studentRepository) {

        this.studentRepository = studentRepository;

    }

    public void showStudent() {

        System.out.println("Student Service Executed");

        studentRepository.displayStudent();

    }

}
```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

       xsi:schemaLocation="
http://www.springframework.org/schema/beans
https://www.springframework.org/schema/beans/spring-beans.xsd">
```

```
<bean id="studentRepository"
      class="com.student.repository.StudentRepository"/>

<bean id="studentService"
      class="com.student.service.StudentService">

    <property name="studentRepository"
              ref="studentRepository"/>

</bean>

</beans>
```

MainApplication.java

```
package com.student;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

import com.student.service.StudentService;

public class MainApplication {

    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        StudentService service =
            (StudentService) context.getBean("studentService");

        service.showStudent();

    }

}
```

Output

Student Service Executed
Student information fetched successfully.

2) Implementing Dependency Injection

StudentRepository.java

```
package com.student.repository;

public class StudentRepository {

    public void getStudentDetails() {

        System.out.println("Retrieving student details from repository...");

    }

}
```

StudentService.java

```
package com.student.service;

import com.student.repository.StudentRepository;

public class StudentService {

    private StudentRepository repository;

    public void setRepository(StudentRepository repository) {

        this.repository = repository;

    }

    public void displayStudentDetails() {

        System.out.println("StudentService is processing...");

        repository.getStudentDetails();

    }

}
```

```
}
```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

       xsi:schemaLocation="
http://www.springframework.org/schema/beans
https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="repository"
          class="com.student.repository.StudentRepository"/>

    <bean id="service"
          class="com.student.service.StudentService">

        <property name="repository"
                  ref="repository"/>

    </bean>

</beans>
```

StudentApplication.java

```
package com.student;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

import com.student.service.StudentService;

public class StudentApplication {

    public static void main(String[] args) {

        ApplicationContext context =
```

```

        new ClassPathXmlApplicationContext("applicationContext.xml");

        StudentService service =
            (StudentService) context.getBean("service");

        service.displayStudentDetails();

    }

}

```

Output

StudentService is processing...
Retrieving student details from repository...

3) Creating and Configuring a Maven Project

pom.xml

```

<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

  xsi:schemaLocation="
    http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.student</groupId>
  <artifactId>StudentManagement</artifactId>
  <version>1.0-SNAPSHOT</version>

  <dependencies>

    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-context</artifactId>
      <version>5.3.30</version>
    </dependency>

    <dependency>
      <groupId>org.springframework</groupId>

```

```
        <artifactId>spring-aop</artifactId>
        <version>5.3.30</version>
    </dependency>

    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-webmvc</artifactId>
        <version>5.3.30</version>
    </dependency>

</dependencies>

<build>

    <plugins>

        <plugin>

            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-compiler-plugin</artifactId>
            <version>3.11.0</version>

            <configuration>

                <source>1.8</source>
                <target>1.8</target>

            </configuration>

        </plugin>

    </plugins>

</build>

</project>
```